A REAL-TIME ADAPTIVE DIGITAL COMMUNICATION SYSTEM FOR INTELLIGENT LINE-CODE SELECTION UNDER SIMULATED UNDERWATER ACOUSTIC CHANNEL CONDITIONS

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MOTIVATION

The problem is not just transmitting data. It is transmitting data reliably when channel conditions change.

- Underwater acoustic communication faces noise, signal attenuation, multipath propagation, and time-varying channel conditions.
- A fixed line-coding scheme cannot necessarily deliver the best performance across every operating condition.
- Different line codes have different bandwidth, spectral, and synchronization characteristics.
- Selecting an appropriate scheme requires evaluating both the channel and the communication requirements.

INTRODUCTION

A software-based communication framework that evaluates multiple line codes and selects a suitable scheme for simulated underwater conditions.

- The framework combines four core functions:
- **Encode**: Generate digital waveforms using NRZ-L, RZ, Manchester, and AMI.
- **Simulate**: Introduce configurable channel impairments.
- **Evaluate**: Analyse BER, PSD, bandwidth, and synchronization characteristics.
- **Adapt**: Apply a decision rule to choose a suitable line code.
- The project brings together fundamental digital communication concepts in a unified, testable framework.

LITERATURE REVIEW

S. No.	Paper name and year	Main research focus	Relation to SignalForge	Research gap
1	Reinforcement Learning-Based Automated Modulation Switching Algorithm for an Enhanced Underwater Acoustic Communication (2024)	• Uses reinforcement learning to switch between FH-BFSK, ASK, PSK and OFDM based on communication conditions.	• Provides a reference for intelligent, automatic signal-scheme selection under simulated or varying underwater conditions.	• Focuses on modulation switching rather than direct selection among line codes such as NRZ, RZ, Manchester and AMI.
2	An Optimal Adaptive Coding and Modulation Mode Selection Method Based on Machine Learning for Underwater Acoustic Communication (2025)	• Uses a random forest classifier and channel quality indicators (SNR, pilot BER, channel impulse response) to select a transmission mode.	• Supports the use of machine learning to select a suitable transmission configuration based on channel conditions.	• Does not specifically investigate adaptive line-code selection across different underwater channel conditions.

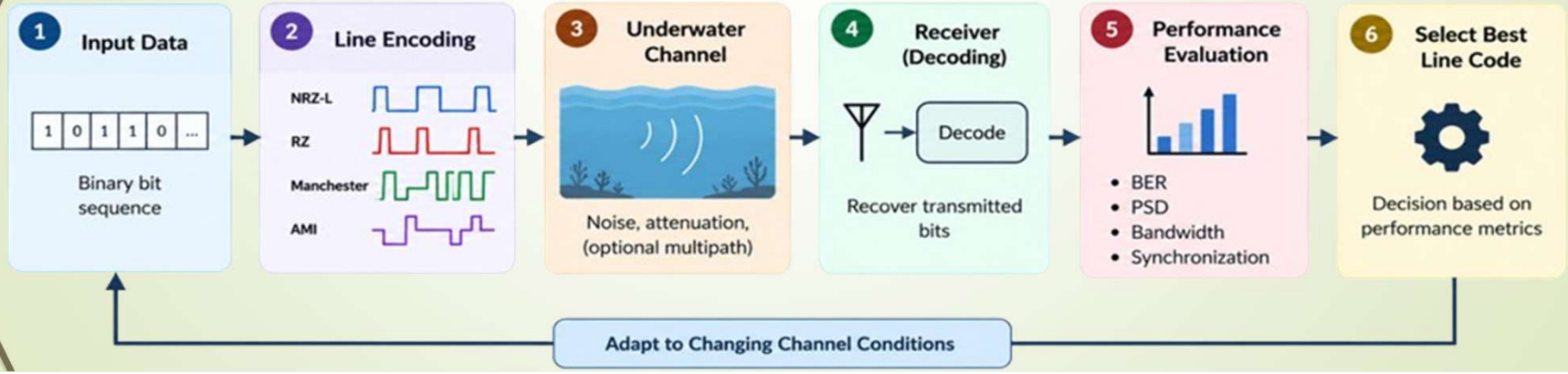
S. No.	Paper name and year	Main research focus	Relation to SignalForge	Research gap
3	Adaptive Modulation and Coding for Underwater Acoustic OTFS Communications Based on Meta-Learning (2024)	• Studies adaptive modulation and coding for underwater acoustic OTFS communications using meta-learning.	• Provides ideas for adapting transmission schemes to rapidly changing underwater channels.	• Focuses on OTFS modulation and coding, not a comparative software framework for conventional line codes.
4	A State-of-the-Art Survey and Benchmarking of Adaptive Modulation and Coding for Underwater Acoustic Communications (2026)	• Reviews and benchmarks conventional, machine-learning and reinforcement-learning approaches to adaptive modulation and coding.	• Helps identify suitable algorithms, evaluation metrics and comparison methods for SignalForge.	• Addresses adaptive modulation and coding broadly; dedicated evaluation of NRZ, RZ, Manchester and AMI selection is not its primary focus.
5	Adaptive Switching for Communication Profiles in Underwater Acoustic Modems Based on Reinforcement Learning (2023)	• Uses deep reinforcement learning to switch communication profiles according to channel conditions, targeting energy and spectral efficiency.	• Provides background for developing an adaptive decision-making algorithm that responds to channel changes.	• Focuses on communication profiles rather than conventional line-code selection.

PROBLEM STATEMENT

- A fixed line-coding scheme is generally chosen in advance, even though its spectral and synchronization characteristics may not suit every communication scenario.
- In simulated underwater acoustic environments, channel impairments can further affect the received signal and the ability to recover transmitted bits.
- **The challenge**: How can a communication framework systematically evaluate competing line codes and select one that best meets the requirements of a changing channel?
- **Our proposed solution**: A software-based adaptive framework that combines channel-condition inputs, line-code performance measurements, and a predefined selection algorithm.

FRAMEWORK

- Generate a binary input sequence.
- Encode the data using NRZ-L, RZ, Manchester, and AMI.
- Pass the encoded signals through simulated underwater channel conditions.
- Decode the received signals and evaluate BER, PSD, bandwidth, and synchronization.
- Apply a predefined decision algorithm to select the most suitable line code.
- Compare its performance against fixed line-coding schemes and reassess when channel conditions change.
- **Expected outcome:** A channel-aware framework for evaluating and selecting suitable line-coding schemes.



RESEARCH OBJECTIVES: What We Aim to Achieve

01. Implement multiple line codes

- Generate and analyse NRZ-L, RZ, Manchester, and AMI using a common input sequence.

02. Model channel conditions

- Simulate configurable noise and attenuation, with additional impairments where supported.

03. Quantify communication performance

- Compare BER, PSD, bandwidth requirements, and synchronization characteristics.

04. Develop adaptive selection logic

- Select a suitable line code according to defined channel and performance criteria.

05. Validate against a fixed baseline

- Test whether the adaptive approach offers a measurable advantage under identical conditions.

NOVELTY: What Makes Our Approach Different?

- **Unified analysis**

Evaluates multiple line codes using both time-domain waveforms and frequency-domain PSD characteristics.

- **Channel-aware selection**

Connects simulated channel conditions and measured performance to an explicit coding-scheme selection rule.

- ***Multi-criteria evaluation***

Considers BER alongside bandwidth, spectral characteristics, and synchronization rather than relying on a single metric.

- **Measurable validation**

Compares adaptive selection with fixed line coding under equivalent simulation conditions.

We don't simply compare line codes, but investigating how performance measurements can guide their selection in a simulated underwater communication environment.

EXPECTED OUTCOMES & FUTURE SCOPE

Expected outcomes

- A software/ML model framework for generating and analysing multiple digital line codes.
- Comparative waveform and PSD visualisations.
- Performance measurements across different simulated channel conditions.
- An adaptive selection mechanism with clearly defined decision criteria.
- A baseline comparison to quantify the benefits and limitations of adaptive selection.

Future scope

- More realistic underwater acoustic channel modelling.
- Evaluation under multipath and Doppler effects.
- Automated decision-rule optimisation.
- Integration with a complete communication link and, eventually, experimental validation.

We plan to build a communication framework that responds to channel conditions rather than relying entirely on a predetermined coding choice.

THANK YOU